Quiz Q1. Let $\mathbf{w} = (1, 3)^\mathsf{T}$, $b = 10$ and $\mathbf{x} = (4, -6)^\mathsf{T}$.

What is the Euclidean distance from $\mathbf{x}$ to the decision boundary?

A. $-2/5$

B. $2/5$

C. $-4/\sqrt{10}$

D. $4/\sqrt{10}$

Quiz Q1 answer. The correct answer is D.

The score and weight norm are

$$f_{\mathbf{w},b}(\mathbf{x}) = 1 \times 4 + 3 \times (-6) + 10 = -4, \qquad \|\mathbf{w}\|_2 = \sqrt{10}.$$

Therefore

$$d(\mathbf{x}, \mathcal{H}) = \frac{|-4|}{\sqrt{10}}.$$

The signed distance is $-4/\sqrt{10}$, but distance itself is nonnegative.

Quiz Q2. In Example 3, the smallest functional margin is 1 and the geometric margin is 5.

Multiply both $\mathbf{w}$ and b by 6. What are the new smallest functional margin and geometric margin, respectively?

A. 6 and 30

B. 6 and 5

C. 1 and 5

D. 1 and 30

Quiz Q2 answer. The correct answer is B.

Scores and functional margins are multiplied by 6.

The weight norm is also multiplied by 6:

$$\min_i m'_i \;=\; 6, \qquad \|\mathbf{w}'\|_2 \;=\; \frac{6}{5}.$$

Hence

$$\gamma' \;=\; \frac{6}{6/5} \;=\; 5.$$

The decision boundary and geometric margin stay unchanged.

Quiz Q3. Keep $\mathbf{w} = (3, 4)^\mathsf{T}/25$ and $b = -2$.

Now give the datapoint $\mathbf{x} = (8, 9)^\mathsf{T}$ label $y = -1$.

What is its hinge loss?

A. 0

B. $3/5$

C. $2/5$

D. $7/5$

Quiz Q3 answer. The correct answer is D.

The score remains

$$f_{\mathbf{w},b}(\mathbf{x}) \;=\; \frac{3\cdot 8+4\cdot 9}{25}-2 \;=\; \frac{2}{5}.$$

But the label changes the functional margin:

$$m \;=\; (-1)\frac{2}{5} \;=\; -\frac{2}{5}, \qquad \ell(m) \;=\; \max\left\{0, 1+\frac{2}{5}\right\} \;=\; \frac{7}{5}.$$

The classifier predicts $+1$, so this negative-labelled point is misclassified.